

Chapter 17

Watchdog Timers (WDT)

17.1 Overview

Watchdog timers are hardware timers used to detect and recover from malfunctions. They must be periodically fed (reset) to prevent a timeout. A system/software that is behaving unexpectedly (e.g., is stuck in a software loop or in overdue events) will fail to feed the watchdog, thus the watchdog timer will time out and trigger an interrupt or reset, directing the system to a known state for exception handling or a restart. Therefore, watchdog timers are useful for detecting and handling erroneous system/software behavior.

As shown in Figure 17.1-1, ESP32-P4 contains three digital watchdog timers: one in each of the two timer groups in Chapter 16 *Timer Group (TIMG)* (called Main System Watchdog Timers, or MWDT) and one in the LP system (called the RTC Watchdog Timer, or RWDT). Each digital watchdog timer allows for four separately configurable stages, and each stage can be programmed to take one action upon timeout, unless the watchdog is fed or disabled. MWDT supports three timeout actions: interrupt, HP CPU reset, and HP core reset, while RWDT supports four timeout actions: interrupt, HP CPU reset, HP core reset, and system reset (see details in Section 17.2.2.2 *Stages and Timeout Actions*). A timeout value can be set for each stage individually.

In SPI Boot mode, RWDT and the MWDT in timer group 0 are enabled automatically in order to detect errors that may occur during the flash boot process and facilitate recovery.

ESP32-P4 also has one analog watchdog timer: Super watchdog (SWD). It is an ultra-low-power circuit in analog domain that helps to prevent the system from operating in a sub-optimal state and resets the system if required.

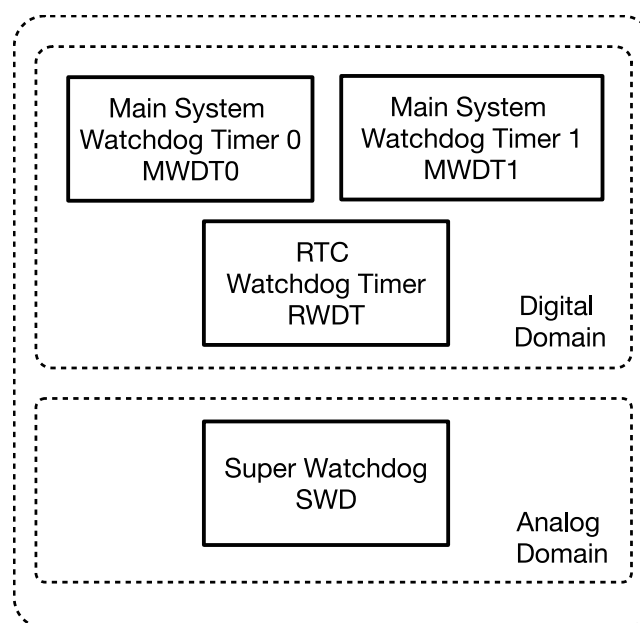


Figure 17.1-1. Watchdog Timers Overview

Note that while this chapter provides the functional descriptions of the watchdog timer's, MWDT register descriptions are detailed in Chapter [16 Timer Group \(TIMG\)](#), and the RWDT and SWD register descriptions are detailed in Section [17.5 Register Summary](#).

17.2 Digital Watchdog Timers

17.2.1 Features

Watchdog timers have the following features:

- Four stages, each with a separately programmable timeout value and timeout action
- Timeout actions:
 - MWDT: interrupt, HP CPU reset, HP core reset
 - RWDT: interrupt, HP CPU reset, HP core reset, system reset
- Flash boot protection under SPI Boot mode at stage 0:
 - MWDT0: HP core reset upon timeout
 - RWDT: system reset upon timeout
- Write protection that makes WDT register read only unless unlocked
- 32-bit timeout counter
- Clock source:
 - MWDT: PLL_F80M_CLK, RC_FAST_CLK or XTAL_CLK
 - RWDT: LP_DYN_SLOW_CLK

17.2.2 Functional Description

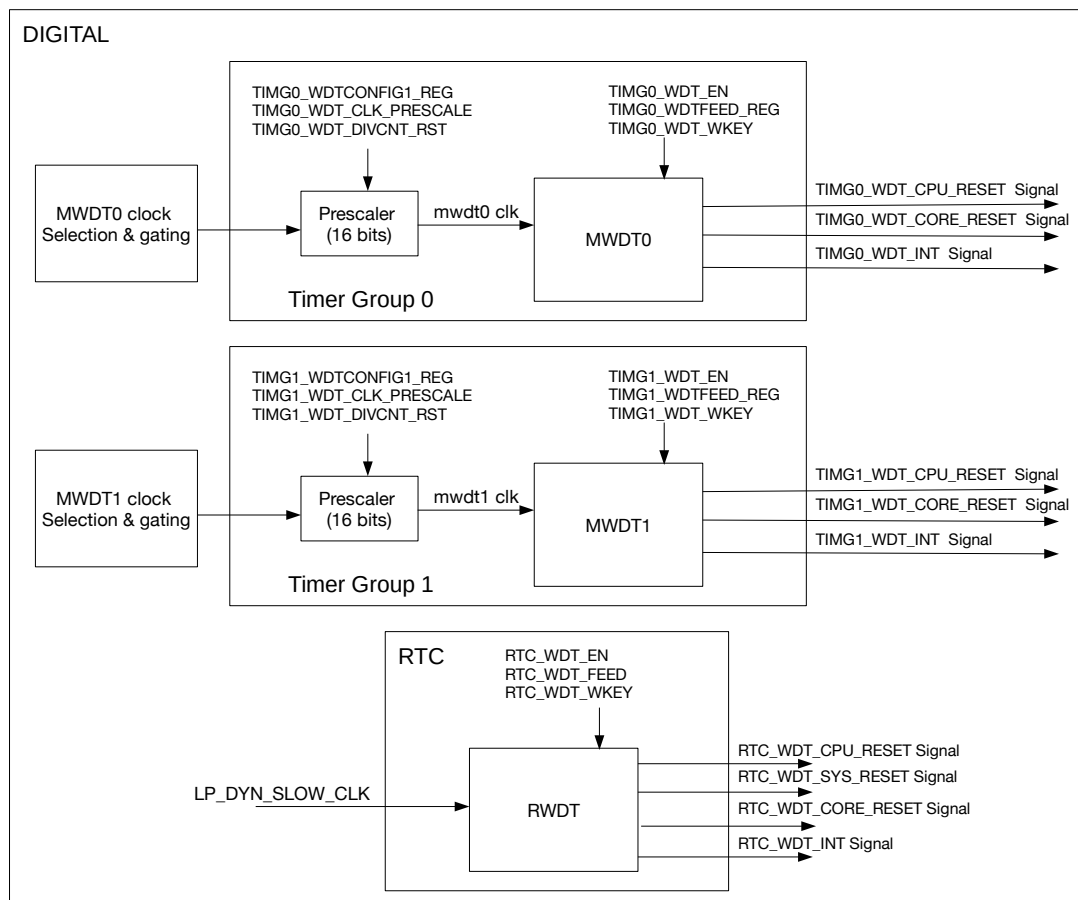


Figure 17.2-1. Digital Watchdog Timers in ESP32-P4

Figure 17.2-1 shows the three watchdog timers in ESP32-P4 digital systems.

17.2.2.1 Clock Source and 32-Bit Counter

At the core of each watchdog timer is a 32-bit counter.

Take MWDTO as an example:

- MWDTO can select between the PLL_F80M_CLK, RC_FAST_CLK or XTAL_CLK (external) clock as its clock source by setting the HP_SYS_TIMERGRPO_WDT_SRC_SEL field of the HP_SYS_CLKRST_PERI_CLK_CTRL20_REG register.
- The selected clock is switched on by setting HP_SYS_TIMERGRPO_WDT_CLK_EN field of the HP_SYS_CLKRST_PERI_CLK_CTRL20_REG register to 1 and switched off by setting it to 0. Then the selected clock is divided by a 16-bit configurable prescaler. See more details in Table 10.2-1 of Chapter 10.
- The 16-bit prescaler for MWDT is configured via the TIMG_WDT_CLK_PRESCALE field of TIMG_WDTCONFIG1_REG. When TIMG_WDT_DIVCNT_RST field is set, the prescaler is reset and it can be re-configured at once.

In contrast, the clock source of RWDT is derived directly from LP_DYN_SLOW_CLK (see details in Chapter 10 [Reset and Clock](#)).

MWDT and RWDT are enabled by setting the [TIMG_WDT_EN](#) and [RTC_WDT_EN](#) fields respectively. When enabled, the 32-bit counters of the watchdog will increment on each source clock cycle until the timeout value of the current stage is reached (i.e., timeout of the current stage). When this occurs, the current counter value is reset to zero and the next stage will become active. If a watchdog timer is fed by software, the timer will return to stage 0 and reset its counter value to zero. Software can feed a watchdog timer by writing any value to [TIMG_WDTFEED_REG](#) for MDWT and by writing 1 to [RTC_WDT_FEED](#) for RWDT.

17.2.2.2 Stages and Timeout Actions

Timer stages allow for a timer to have a series of different timeout values and corresponding timeout action. When one stage times out, the timeout action is triggered, the counter value is reset to zero, and the next stage becomes active.

MWDT/RWDT offers four stages (referred to as stages 0 to 3). The watchdog timers will progress through each stage in a loop (i.e., from stage 0 to 3, then back to stage 0).

Timeout values of each stage for MWDT are configured in [TIMG_WDTCONFIG_i_REG](#) (where *i* ranges from 2 to 5), whilst timeout values for RWDT are configured using [RTC_WDT_STG_j_HOLD](#) field (where *j* ranges from 0 to 3).

Please note that the timeout value of stage 0 for RWDT (T_{hold0}) is determined by the combination of the

[EFUSE_WDT_DELAY_SEL](#) field of eFuse register [EFUSE_RD_REPEAT_DATA0_REG](#) and [RTC_WDT_STG0_HOLD](#) field. The relationship is as follows:

$$T_{hold0} = \text{RTC_WDT_STG0_HOLD} \ll (\text{EFUSE_WDT_DELAY_SEL} + 1)$$

where $\ll$ is a left-shift operator. For example, if [RTC_WDT_STG0_HOLD](#) is configured as 100 and [EFUSE_WDT_DELAY_SEL](#) is 1, the T_{hold0} will be 400 cycles.

Upon the timeout of each stage, one of the following timeout actions will be executed:

Table 17.2-1. Timeout Actions

Timeout Action	Description
Interrupt	Trigger an interrupt
HP CPU reset	Reset the HP CPU0 and HP CPU1
HP core reset	HP core reset resets HP CPU0, HP CPU1, HP peripherals, HP GPIO, etc.
System reset	Reset the whole digital system, including the LP system.
Disabled	No effect on the system

For MWDT, the timeout action of all stages is configured in [TIMG_WDTCONFIG0_REG](#). Likewise for RWDT, the timeout action is configured in [RTC_WDT_CONFIG0_REG](#).

17.2.2.3 Write Protection

Watchdog timers are critical to detecting and handling erroneous system/software behavior, thus should not be disabled easily (e.g., due to a misplaced register write). Therefore, MWDT and RWDT incorporate a write protection mechanism that prevent the watchdogs from being disabled or tampered with due to an accidental write.

The write protection mechanism is implemented using a write-key field for each timer ([TIMG_WDT_WKEY](#) for MWDT, [RTC_WDT_WKEY](#) for RWDT). The value 0x50D83AA1 must be written to the watchdog timer's write-key field before any other register of the same watchdog timer can be changed. Any attempts to write to a watchdog timer's registers (other than the write-key field itself) whilst the write-key field's value is not 0x50D83AA1 will be ignored. The recommended procedure for accessing a watchdog timer is as follows:

1. Disable the write protection by writing the value 0x50D83AA1 to the timer's write-key field.
2. Make the required modification of the watchdog such as feeding or changing its configuration.
3. Re-enable write protection by writing any value other than 0x50D83AA1 to the timer's write-key field.

17.2.2.4 Flash Boot Protection

Under the SPI boot mode, MWDT0 as well as RWDT, are automatically enabled. Stage 0 for the enabled MWDT0 is automatically configured as HP core reset action upon timeout, known as HP core reset. Likewise, stage 0 for RWDT is configured to system reset, which resets the main system and RTC when it times out. After SPI booting, [TIMG_WDT_FLASHBOOT_MOD_EN](#) and [RTC_WDT_FLASHBOOT_MOD_EN](#) should be cleared to stop the flash boot protection procedure for both MWDT0 and RWDT respectively. After this, MWDT0 and RWDT can be configured by software.

17.3 Super Watchdog

Super watchdog (SWD) is an ultra-low-power circuit in analog domain that helps to prevent the system from operating in a sub-optimal state and resets the system (system reset) if required. SWD contains a watchdog circuit that needs to be fed for at least once during its timeout period, which is slightly less than one second. About 100 ms before watchdog timeout, it will also send out a WD_INTR signal as a request to remind the system to feed the watchdog.

If the system doesn't respond to SWD feed request and watchdog finally times out, SWD will generate a system level signal SWD_RSTB to reset whole digital circuits on the chip (system reset) .

The source of the clock for SWD is constant and can not be selected.

17.3.1 Features

SWD has the following features:

- Ultra-low power
- Interrupt to indicate that the SWD is about to time out
- Various dedicated methods for software to feed SWD, which enables SWD to monitor the working state of the whole operating system

17.3.2 Super Watchdog Controller

17.3.2.1 Structure

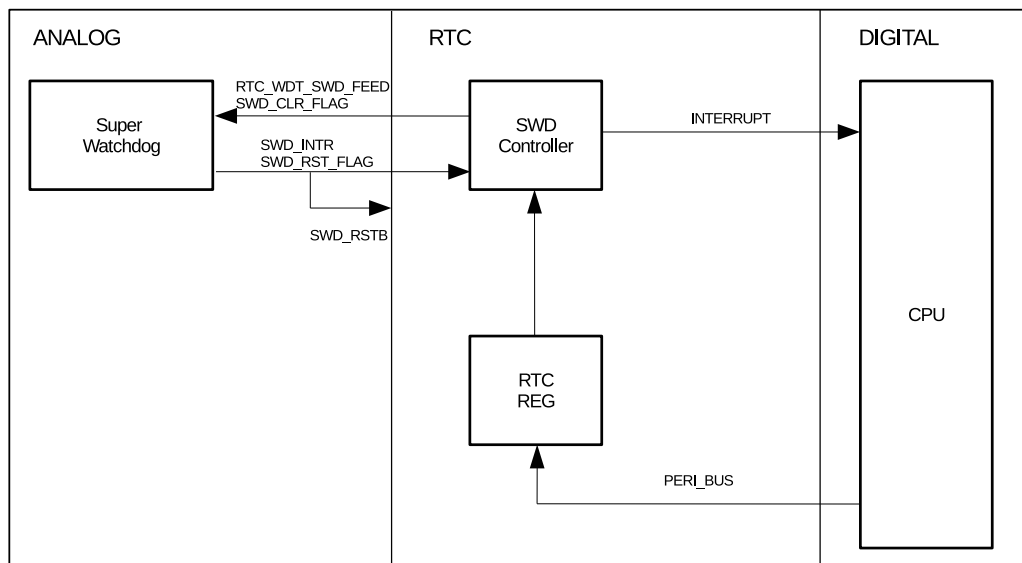


Figure 17.3-1. Super Watchdog Controller Structure

17.3.2.2 Workflow

In normal state:

- SWD controller receives feed request from SWD.
- SWD controller can send an interrupt to main CPU.
- Main CPU can feed SWD directly by setting `RTC_WDT_SWD_FEED`.
- When trying to feed SWD, CPU needs to disable SWD controller's write protection by writing `0x50D83AA1` to `RTC_WDT_SWD_WKEY`. This prevents SWD from being fed by mistake when the system is operating in sub-optimal state.
- If setting `RTC_WDT_SWD_AUTO_FEED_EN` to 1, SWD controller can also feed SWD itself without any interaction with CPU.

After reset:

- Check `LP_AONCLKRST_LPCORE_RESET_CAUSE[4:0]` for the cause of HP CPU reset.
If `LP_AONCLKRST_LPCORE_RESET_CAUSE[4:0] == 0x12`, it indicates that the cause is SWD reset.
- Set `RTC_WDT_SWD_RST_FLAG_CLR` to clear the SWD reset flag.

17.4 Interrupts

For watchdog timer interrupts, please refer to Section [16.3.7 Interrupts](#) in Chapter [16 Timer Group \(TIMG\)](#).

17.5 Register Summary

The addresses in this section are relative to RTC_WDT base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

Name	Description	Address	Access
configuration register			
RTC_WDT_CONFIG0_REG	Configure the RWDT operation	0x0000	R/W
RTC_WDT_CONFIG1_REG	Configure the RWDT timeout time of stage0	0x0004	R/W
RTC_WDT_CONFIG2_REG	Configure the RWDT timeout time of stage1	0x0008	R/W
RTC_WDT_CONFIG3_REG	Configure the RWDT timeout time of stage2	0x000C	R/W
RTC_WDT_CONFIG4_REG	Configure the RWDT timeout time of stage3	0x0010	R/W
RTC_WDT_FEED_REG	Configure the feed function of RWDT	0x0014	WT
RTC_WDT_WPROTECT_REG	Configure the lock function of RWDT	0x0018	R/W
RTC_WDT_SWD_CONFIG_REG	Configure the SWD operation	0x001C	varies
RTC_WDT_SWD_WPROTECT_REG	Configure the lock function of SWD	0x0020	R/W
RTC_WDT_INT_RAW_REG	Configure whether to generate timeout interrupt	0x0024	R/WTC/SS
RTC_WDT_INT_ST_REG	The interrupt status register of WDT	0x0028	RO
RTC_WDT_INT_ENA_REG	The interrupt enable register of WDT	0x002C	R/W
RTC_WDT_INT_CLR_REG	The interrupt clear register of WDT	0x0030	WT
RTC_WDT_DATE_REG	Version control register	0x03FC	R/W

17.6 Registers

MWDT registers are part of the timer submodule and are described in Section 16.5 *Register Summary* in Chapter 16 *Timer Group (TIMG)*.

The addresses of RWDT and SWD registers in this section are relative to RTC_WDT base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

Register 17.1. RTC_WDT_CONFIG0_REG (0x0000)

RTC_WDT_EN		RTC_WDT_STG0		RTC_WDT_STG1		RTC_WDT_STG2		RTC_WDT_STG3		RTC_WDT_CPU_RESET_LENGTH		RTC_WDT_SYS_RESET_LENGTH		RTC_WDT_FLASHBOOT_MOD_EN		RTC_WDT_PROCPU_RESET_EN		RTC_WDT_PAUSE_IN_SLP		(reserved)		(reserved)										
31	30	28	27	25	24	22	21	19	18	16	15	13	12	11	10	9	8															0
0	0x0		0x0		0x0		0x0		0x1		0x1		1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset		

RTC_WDT_PAUSE_IN_SLP Configure whether or not pause RWDT when chip is in sleep mode.

0: Enable

1: Disable

(R/W)

RTC_WDT_PROCPU_RESET_EN Configure whether or not to enable RWDT to reset CPU.

0: Disable

1: Enable

(R/W)

RTC_WDT_FLASHBOOT_MOD_EN Configure whether or not to enable RWDT when chip is in SPI boot mode.

0: Disable

1: Enable

(R/W)

RTC_WDT_SYS_RESET_LENGTH Configure the HP core reset time.

Measurement unit: LP_DYN_FAST_CLK

(R/W)

RTC_WDT_CPU_RESET_LENGTH Configure the HP CPU reset time.

Measurement unit: LP_DYN_FAST_CLK

(R/W)

RTC_WDT_STG3 Configure the timeout action of stage3.

0: No operation

1: Generate interrupt

2: Generate HP CPU reset

3: Generate HP core reset

4: Generate system reset

(R/W)

Continued on the next page...

Register 17.1. RTC_WDT_CONFIG0_REG (0x0000)**Continued from the previous page...****RTC_WDT_STG2** Configure the timeout action of stage2.

- 0: No operation
- 1: Generate interrupt
- 2: Generate HP CPU reset
- 3: Generate HP core reset
- 4: Generate system reset
- (R/W)

RTC_WDT_STG1 Configure the timeout action of stage1.

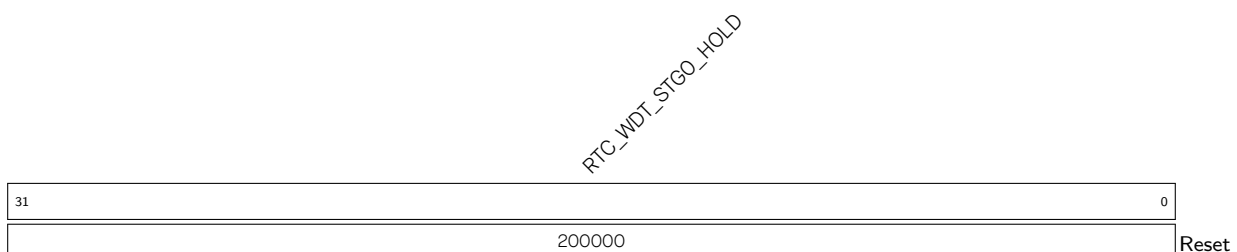
- 0: No operation
- 1: Generate interrupt
- 2: Generate HP CPU reset
- 3: Generate HP core reset
- 4: Generate system reset
- (R/W)

RTC_WDT_STG0 Configure the timeout action of stage0.

- 0: No operation
- 1: Generate interrupt
- 2: Generate HP CPU reset
- 3: Generate HP core reset
- 4: Generate system reset
- (R/W)

RTC_WDT_EN Configure whether or not enable RWDT.

- 0: Disable RWDT
- 1: Enable RWDT
- (R/W)

Register 17.2. RTC_WDT_CONFIG1_REG (0x0004)**RTC_WDT_STGO_HOLD** Configure the timeout time for stage0.

- Measurement unit: LP_DYN_SLOW_CLK
- (R/W)

Register 17.3. RTC_WDT_CONFIG2_REG (0x0008)

RTC_WDT_STG1_HOLD

31	0
80000	
Reset	

RTC_WDT_STG1_HOLD Configure the timeout time for stage1.

Measurement unit: LP_DYN_SLOW_CLK

(R/W)

Register 17.4. RTC_WDT_CONFIG3_REG (0x000C)

RTC_WDT_STG2_HOLD

31	0
0x000fff	
Reset	

RTC_WDT_STG2_HOLD Configure the timeout time for stage2.

Measurement unit: LP_DYN_SLOW_CLK

(R/W)

Register 17.5. RTC_WDT_CONFIG4_REG (0x0010)

RTC_WDT_STG3_HOLD

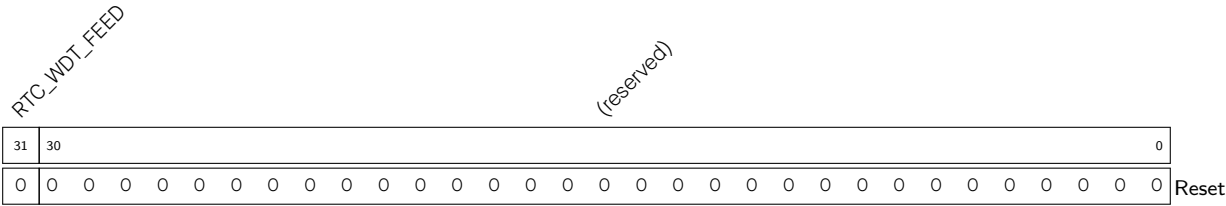
31	0
0x000fff	
Reset	

RTC_WDT_STG3_HOLD Configure the timeout time for stage3.

Measurement unit: LP_DYN_SLOW_CLK

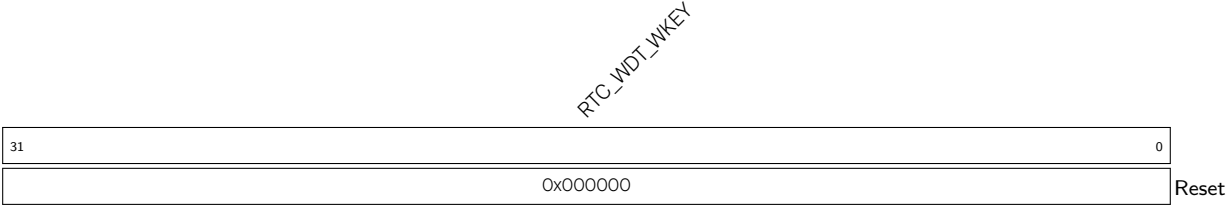
(R/W)

Register 17.6. RTC_WDT_FEED_REG (0x0014)



RTC_WDT_FEED Configure this bit to feed the RWDT.
0: Invalid
1: Feed RWDT
(WT)

Register 17.7. RTC_WDT_WPROTECT_REG (0x0018)



RTC_WDT_WKEY Configure this field to lock or unlock RWDT's configuration registers.
0x50D83AA1: unlock the RWDT configuration register
Others value: lock the RWDT configuration register which can't be modified by software.
(R/W)

Register 17.8. RTC_WDT_SWD_CONFIG_REG (0x001C)

RTC_WDT_SWD_FEED RTC_WDT_SWD_DISABLE			RTC_WDT_SWD_SIGNAL_WIDTH							RTC_WDT_SWD_RST_FLAG_CLR RTC_WDT_SWD_AUTO_FEED_EN										(reserved)										RTC_WDT_SWD_RESET_FLAG	
31	30	29	20							19	18	17													1	0					
0	0	300							0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

RTC_WDT_SWD_RESET_FLAG Represents the SWD whether has generated the reset signal

0: No

1: Yes

(RO)

RTC_WDT_SWD_AUTO_FEED_EN Configure this bit to enable to feed SWD automatically by hardware.

0: Disable

1: Enable

(R/W)

RTC_WDT_SWD_RST_FLAG_CLR Configure this bit to clear SWD reset flag

0: Invalid

1: Clear the reset flag

(WT)

RTC_WDT_SWD_SIGNAL_WIDTH Configure the SWD signal length that output to analog circuit.

Measurement unit: LP_DYN_FAST_CLK (R/W)

RTC_WDT_SWD_DISABLE Configure this bit to disable the SWD.

0: Enable the SWD

1: Disable the SWD

(R/W)

RTC_WDT_SWD_FEED Configure this bit to feed the SWD.

0: Invalid

1: Feed SWD

(WT)

RTC_WDT_SWD_WKEY

RTC_WDT_SWD_WKEY Configure this field to lock or unlock SWD's configuration registers.
0x50D83AA1: unlock the SWD configuration register.
Others value: lock the SWD configuration register which can't be modified by the software.
(R/W)

RTC_WDT_SWDT_INT_RAW
RTC_WDT_SWDT_INT_RAW

(reserved)

RTC_WDT_RWDT_INT_RAW Represents the RWDT whether or not generates timeout interrupt.
0: No
1: Yes
(R/WTC/SS)

Register 17.11. RTC_WDT_INT_ST_REG (0x0028)

Diagram illustrating the structure of the **RTC_WDTR** register (32 bits wide):

- Bits 31 and 30: **RTC_WDT_RWDT_INT_ST**
- Bit 29: **RTC_WDT_SWDT_INT_ST**
- Bits 0 to 28: **(reserved)**

RTC_WDT_SWDT_INT_ST Represents the SWD whether or not has generated and sent timeout interrupt to CPU.

0: No

1: Yes

(RO)

RTC_WDT_RWDT_INT_ST Represents the RWDT whether or not generates and sends timeout interrupt to CPU.

0: No

1: Yes

(RO)

Register 17.12. RTC_WDT_INT_ENA_REG (0x002C)

Diagram of the 32-bit SWD_TDR register structure:

- Bits 31-30: RTC_WDT_RWDT_INT_ENA (2 bits)
- Bits 29-0: RTC_WDT_SWDT_INT_ENA (30 bits)
- Bits 31-0: (reserved) (32 bits)

The register is shown as a horizontal bar with bit positions 31, 30, and 29 marked. The value 0 is shown in the reserved field.

RTC_WDT_SWDT_INT_ENA Configure whether or not to enable the SWD to send timeout interrupt.

0: Disable

1: Enable

(R/W)

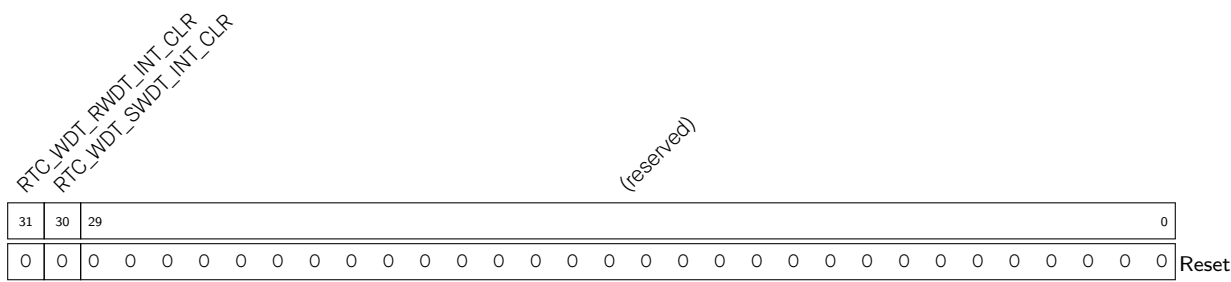
RTC_WDT_RWDT_INT_ENA Configure whether or not to enable the RWDT to send timeout interrupt.

0: Disable

1: Enable

(R/W)

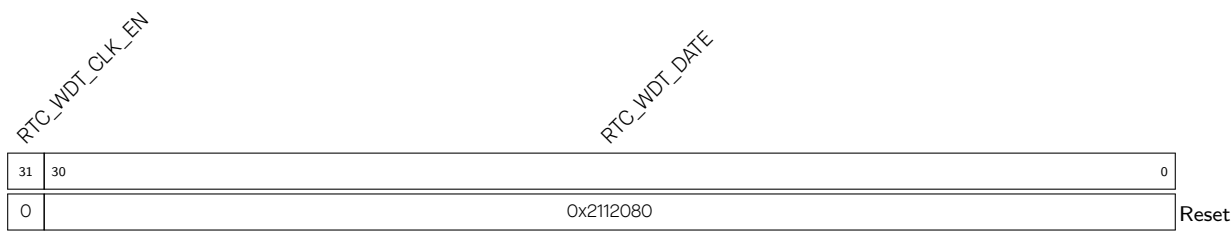
Register 17.13. RTC_WDT_INT_CLR_REG (0x0030)



RTC_WDT_SWDT_INT_CLR Configure whether to clear the timeout interrupt signal sent by SWD to CPU.
0: No
1: Yes
(WT)

RTC_WDT_RWDT_INT_CLR Configure whether to clear the timeout interrupt signal sent by RWDT to CPU.
0: No
1: Yes
(WT)

Register 17.14. RTC_WDT_DATE_REG (0x03FC)



RTC_WDT_DATE Version control register. (R/W)
RTC_WDT_CLK_EN Reserved. (R/W)